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cap program drop skdecomp
program define skdecomp, rclass sortpreserve byable(recall)
    version 10.0, missing
    if c(more)=="on" set more off
    local version : di "version " string(_caller()) ", missing:"
    syntax varlist(numeric min=1 max=1) [if] [in] [aweight fweight], by(varname
numeric)    ///
                varpl(varname numeric)
    ///
                [mpl(numlist sort) INdicator(string) idpl(varname)]

    tempname a b temp PATH A B C D pov_rate
    tempvar varname2
    marksample touse, strok

    preserve

    * * Indicator
    if ("`indicator'"=="") {
        local fgt0 "fgt0"
    }
    else {
        local t 0
        forvalues i = 0(1)2 {
            local t = regexm("`indicator'", "fgt`i'") + `t'
        }
        if `t'==0 {
            di in red "Indicator not valid"
            exit 198
        }
        if regexm("`indicator'", "fgt0")!=0 local fgt0 "fgt0"
        if regexm("`indicator'", "fgt1")!=0 local fgt1 "fgt1"
        if regexm("`indicator'", "fgt2")!=0 local fgt2 "fgt2"
    }

    * ** Weights
    if ("`weight'"!="") {
        local weight "[`weight'`exp']"
        local wvar : word 2 of `exp'
    }

    qui markout `touse' `varlist' `wvar' `by' `varpl' `idpl', strok
    qui keep if `touse'

    * ** Comparison variable
    cap tab `by', matrow(`temp')
    local ct = r(N)
    if r(N)==0 error 2000
    if (`r(r)'~2)|(_rc==134) {
        di in red "Only 2 groups allow"
        exit 198
    }
    local c = `temp'[1,1]
    local d = `temp'[2,1]

    * ** Poverty Line Check
    sum `varpl' if `by'==`c', meanonly

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local testc1 = round(`r(max)', .01)
local testc2 = round(`r(min)', .01)
sum `varpl' if `by'==`d', meanonly
local testd1 = round(`r(max)', .01)
local testd2 = round(`r(min)', .01)
if (( `testc1' != `testc2') | ( `testd1' != `testd2')) & ( "`idpl'"=="") {
    di in red "Poverty line is not constant in each `by'. Use the option IDPL"
    exit 198
}

local ing1 "`varlist'"
local ing2 "`varname2'"

* Generate the variables that has the same distribution, but have the average from
another year
qui gen double `ing2' = .

qui sum `ing1' `weight' if `by'==`c' , meanonly
local m1 `r(mean)'

qui sum `ing1' `weight' if `by'==`d' , meanonly
local m2 `r(mean)'

qui replace `ing2'=`ing1'*(`m2'/`m1') if `by'==`c'
qui replace `ing2'=`ing1'*(`m1'/`m2') if `by'==`d'

* Poverty line by period
if "`idpl'"!="" {
    tempvar varpl1
    qui transpose_pline, plvar(`varpl') by(`by') strata(`idpl')
generate(`varpl1')

    * Poverty Line in t=0
    qui gen double lp1 =`varpl' if `by'==`c'
    qui gen double lp2 =`varpl1' if `by'==`c'

    * Poverty Line in t=1
    qui replace lp1 =`varpl1' if `by'==`d'
    qui replace lp2 =`varpl' if `by'==`d'
}
else {
    * Poverty Line in t=0
    qui sum `varpl' if `by'==`c', meanonly
    local lp1 `r(mean)'
    gen double lp1=`lp1'

    * Poverty Line in t=1
    qui sum `varpl' if `by'==`d', meanonly
    local lp2 `r(mean)'
    gen double lp2=`lp2'
}

***=== Shorrocks-Kolenikov Decomposition ===***

foreach year in `c' `d' {
    foreach income in `ing1' `ing2' {
        foreach povline in lp1 lp2 {
            _mpl `income' `weight' if (`by'==`year') ,
varpl(`povline') mpl(`mpl') in(`fgt0' `fgt1' `fgt2')
            mat `a' = r(b)
            mat `income' `_year' `povline' = `a'[1...,3]
        }
    }
}

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* ** Saving results

* poverty line
mat `C' = `a'[1...,1]
* indicator
mat `D' = `a'[1...,2]

mat `A' = (`ing1'`c'_lp1, `ing1'`c'_lp2, `ing2'`c'_lp1, `ing2'`c'_lp2)
mat `B' = (`ing1'`d'_lp2, `ing1'`d'_lp1, `ing2'`d'_lp2, `ing2'`d'_lp1)

** Matrix of the three effects: Growth, Distribution and Poverty Line respectively
mat `PATH' = /*
Growth effect
*/(J(rowsof(`A'),1,1), `D', `C', `B'[1...,2]-`B'[1...,4], `A'[1...,3]-`A'[1...,1],
`A'[1...,3]-`A'[1...,1], `A'[1...,4]-`A'[1...,2], `B'[1...,1]-`B'[1...,3], `B'[1...,1]-
`B'[1...,3] \ /*
Distribution effect
*/ J(rowsof(`A'),1,2), `D', `C', `B'[1...,4]-`A'[1...,1], `B'[1...,2]-`A'[1...,3],
`B'[1...,1]-`A'[1...,4], `B'[1...,1]-`A'[1...,4], `B'[1...,3]-`A'[1...,2], `B'[1...,4]-
`A'[1...,1] \ /*
Prices effect
*/ J(rowsof(`A'),1,3), `D', `C', `B'[1...,1]-`B'[1...,2], `B'[1...,1]-`B'[1...,2],
`A'[1...,4]-`A'[1...,3], `A'[1...,2]-`A'[1...,1], `A'[1...,2]-`A'[1...,1], `B'[1...,3]-
`B'[1...,4] \ /*
Total change
*/ J(rowsof(`A'),1,4), `D', `C', `B'[1...,1]-`A'[1...,1], `B'[1...,1]-`A'[1...,1],
`B'[1...,1]-`A'[1...,1], `B'[1...,1]-`A'[1...,1], `B'[1...,1]-`A'[1...,1], `B'[1...,1]-
`A'[1...,1])

** Poverty rates by period
mat `pov_rate' = (J(rowsof(`A'),1,0) ,`C', `D', `A'[1...,1]) \
(J(rowsof(`A'),1,1), `C', `D', `B'[1...,1]) \ (J(rowsof(`A'),1,2), `C', `D', `B'[1...,1]-
`A'[1...,1])
mat colnames `pov_rate' = Period Line Indicator rate
cap drop _all
cap svmat double `pov_rate' , n(col)

p.p."
label define Period 0 "Rate in `by' `c'" 1 "Rate in `by' `d'" 2 "Total change in
label values Period Period
label var Period "Poverty rates"

if "`mpl'"!="" {
    local pl "Line"
    label var `pl' "By multiples of `varpl'"
    cap tab `pl'
    local m = `r(r)'
    forvalues i =1(1)`m' {
        local j : word `i' of `mpl'
        label define `pl' `i' "`j'", add
    }
    label values `pl' `pl'
}

label define Indicator 0 "FGT0" 1 "FGT1" 2 "FGT2"
label values Indicator Indicator

display as text _newline "{hline 45}"

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display as text "Poverty rates"
display as text "{hline 45}"
tabdisp Period Indicator, cell(rate) format(%12.2fc) by(`pl')

mat `pov_rate' = (J(rowsof(`A'),1,`c'),`D',`A'[1...,1],`C') \
(J(rowsof(`A'),1,`d'),`D',`B'[1...,1],`C')
mat colnames `pov_rate' = Period Indicator Rate Line
return matrix poverty = `pov_rate'

** Results decomposition
mat colnames `PATH' = Effect Indicator Line effect1 effect2 effect3 effect4
effect5 effect6
cap drop _all
cap svmat double `PATH', n(col)

label values Indicator Indicator

label define Effect 1 "Growth" 2 "Redistribution" 3 "Line" 4 "Total change in
p.p." label values Effect Effect

egen effect_avg=rowmean(effect1 effect2 effect3 effect4 effect5 effect6)

if "`mpl'"!="" {
    label var `pl' "By multiples of `varpl'"
    label values `pl' `pl'
}

di
display as text _newline "{hline 55}"
display as text in yellow "Growth, Distribution and Line - Poverty Decomposition"
display as text "{hline 55}"
di as txt "Welfare variable      : `varlist'"
di as txt "Comparison variable : `by'"
di as txt "Number of obs        : "as res %8.0f `ct'

tabdisp Effect Indicator, cell(effect_avg) format(%12.2fc) by(`pl')

sort `pl' Indicator Effect, stable
mkmat `pl' Indicator Effect effect1 effect2 effect3 effect4 effect5 effect6
effect_avg, matrix(`a')
mkmat `pl' Indicator Effect effect_avg, matrix(`b')

return matrix shapley = `a'
return matrix b = `b'

clear

end

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* ** Another program need to calculate poverty
cap program drop _mpl
program define _mpl, rclass sortpreserve byable(recall)
    version 10.0, missing
    if c(more)=="on" set more off
    local version : di "version " string(_caller()) ", missing:"

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syntax varlist(numeric min=1 max=1) [if] [in] [aweight fweight], [varpl(varname
numeric) LINEs(numlist sort) mpl(numlist sort) INdicator(string) max]

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tempname fgt
tempvar w wwvar point0 point1 aux
marksample touse

qui {

    * * Indicator
    if regexm("`indicator'", "fgt0") != 0 local fgt0 "fgt0"
    if regexm("`indicator'", "fgt1") != 0 local fgt1 "fgt1"
    if regexm("`indicator'", "fgt2") != 0 local fgt2 "fgt2"

    * * Weight variable
    if ("`weight'"=="") {
        gen `w' = 1
        local wvar "`w'"
    }
    else {
        local weight "[`weight'`exp']"
        local wvar : word 2 of `exp'
    }
    sum `wvar' if `touse' , meanonly
    local pop_sum = r(sum)
    local t_obs = r(N)
    gen double `wwvar' = `wvar' / `pop_sum'

    * * Points
    sum `varlist' if `touse' , meanonly
    local max1 `r(max)'
    local min `r(min)'
    if `min' > 0 local min 0

    if ("`max'"=="") local max1 ""

    * * varpl
    if ("`varpl'"=="") {
        tempvar varpl
        gen byte `varpl' = 1 if `touse'
        local mpl "`lines'"
    }
    if ("`mpl'"=="") local mpl "1"

    gen double `point0' = .
    gen double `point1' = .
    gen double `aux' = .
    local list "`min' `mpl'"

    local signal "<"

    local cont 0
    foreach pt1 in `mpl' `max1' {

        local ++cont
        local pt0 : word `cont' of `list'

        replace `point0' = `pt0'*`varpl'
        replace `point1' = `pt1'*`varpl'

        if "`pt0'"=="`min'" {
            replace `point0' = `min'
        }
    }
}

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if      "`pt1'"=="`max1'" {
    replace `point1' = `max1'
    local signal "<="
}

* Observation on range
if (`point0'>`point1') {
    di in red "The range used not valid" _new
    exit 198
}

/* Indicators Calculation */

* ** fgt0

tempvar in
gen double `in' = (`varlist'>=`point0'-0.00000001) &
`varlist'`signal'(`point1')) if `touse'
if("`fgt0'"~="") {
    sum `wvar' if `touse' & `in'==1, meanonly
    local pop_sum_in `r(sum)'
    local fgt0 = 100*`pop_sum_in'/'pop_sum'

    mat `fgt' = nullmat(`fgt') \ (`cont', 0, `fgt0')
}
* ** fgt1
if("`fgt1'"~="") {
    replace `aux' = 100*`in'*`wwvar'*(((`point1')-
`varlist')/(`point1')) if `touse'
    sum `aux' if `touse' , meanonly
    local fgt1 `r(sum)'

    mat `fgt' = nullmat(`fgt') \ (`cont', 1, `fgt1')
}

* ** fgt2
if("`fgt2'"~="") {
    replace `aux' = 100*`in'*`wwvar'*((((`point1')-
`varlist')/(`point1'))^2) if `touse'
    sum `aux' if `touse' , meanonly
    local fgt2 `r(sum)'

    mat `fgt' = nullmat(`fgt') \ (`cont', 2, `fgt2')
}

}

return matrix b = `fgt'

end

* ** Another program need to switch poverty lines between periods
cap program drop transpose_pline
program define transpose_pline, nclass
    syntax, plvar(varname numeric) by(varname numeric) [strata(string)]
    generate(string)

    tempfile temp1 temp2

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tempvar univ

su `by', meanonly
local obs `r(N)'
local univ1 `r(min)'
local univ2 `r(max)'

sort `by' `strata' , stable
save `temp1'.dta, replace

* organizing data to switch between two periods
gen `univ' = `univ2' if `by'==`univ1'
replace `univ' = `univ1' if `by'==`univ2'
keep `univ' `strata' `plvar'
duplicates drop `univ' `strata' `plvar', force
rename `plvar' `generate'
sort `univ' `strata' , stable
save `temp2'.dta, replace

use `temp1'.dta, replace
gen `univ' = `by'
sort `univ' `strata' , stable
merge `univ' `strata' using `temp2'.dta, nokeep
cap sum _merge if _merge==1, meanonly
if `r(N)'!=0 di in red "Warning: `r(N)' observations were not matched. Please,
revise the identifier for poverty line, `strata'"
cap sum _merge if _merge==3, meanonly
if `r(N)'!=`obs' di in red "Warning: `strata' is not unique identifier for
poverty line"
drop _merge `univ'

end

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